

TO DO [ITR1]

- USER STORIES
 - **Done Part Way Incomplete**
 - Each Tab Functionality and menu is functional
 - User Login
 - User Authentication
 - Remember Me and Sign-Ups
 - Store recent searches in memory
 - Create a History Test for methods on how to solve
 - Display YouTube Video
 - Display google searches
 - **Need to limit searches**

Meeting Minutes per ITR1

- Feb 3rd: Met for 90 minutes
- Feb 10th: Met for 90 minutes
- Feb 13th: Talked digitally for 45 minutes

Development Tasks Per User Story & Task Assignment

User Story	Development Task	Team Member	Time Allocated	Time Taken
Each Tab Functional	Create menu framework	Otokpa Attah Pietro Angiolella	1 day	0.5 days
Each Tab Functional	Streamline tabs and make each tab similar to the UI	Pietro Angiolella	0.5 days	0.5 days
User Authentication	Create Login Authentication <->Google Firestore	Rishab Yadav	2 days	2 days
Each Tab Functional	Created user Interface for the website using Bootstrap elements & added media tags for cross-device compatibility.	Rishab Yadav	3 days	3 days
User Authentication	Create Register User Database, take input from HTML-> Google Firestore	Rishab Yadav	2 days	2 day
Each Tab Functional	Created Unit Test Cases for Firebase Authentication server, YouTube Search and Google Search.	Rishab Yadav	2 days	2 days
Store User History	Make history store in memory	Pietro Angiolella	2 days	3 days

Store User History	Make history display	Pietro Angiolella	2 days	4 days
Display External Sources	Created custom Google search API key & created JS file to display	Waleed	1 day	1 day
Display External Sources	Create Youtube search API key & created JS file to display	Rishab Yadav	1 day	1 day
Store User History	Used cookies to store user search history	Ondrea Peters	0.5 days	0.5 days

Rationale Behind Changes

1. Google Searches

- We have decided to incorporate Google searches along with YouTube as sources of information to enhance roadmaps while minimizing the risk of accidental plagiarism(All sources will be listed, and links will be provided).
- The most significant implemented feature remains the integration of YouTube videos and Google searches, specifically targeting **PDF articles** for educational purposes. Currently, the system displays the most relevant YouTube video, along with key details such as likes, description, and channel information.

2. Search Course Overhaul

- We have made several improvements to course searches, moving beyond just displaying the course time and professor. Now, using an AI-powered web scraping tool, we plan to provide a **university-specific outline** of each course for more detailed insights.
- To support this enhancement, we have added a **University** input field to the **Create Account** page. This information is stored alongside other user data in **Google Firestore**, allowing us to expand on features in the future. With this, we can track search history and other relevant fields to offer a more **personalized** and **tailored** search experience.

3. Quiz Generation Changes

- We've made some changes to quiz generation. Now, you can implement different question styles and select multiple topics to test on instead of just one.
- Taking flashcard quizzes on multiple topics at once enhances learning by improving memory retention, cognitive flexibility, and problem-solving skills.

Mixing subjects prevents boredom, reduces mental fatigue, and strengthens recall through interleaved learning, making it easier to distinguish concepts. It also helps in recognizing cross-topic connections, which is valuable for real-world applications. Additionally, spaced repetition across different subjects reinforces long-term memory better than focusing on just one. This approach prepares you for exams and complex problem-solving, where diverse knowledge is often required.

(<https://www.scientificamerican.com/article/the-interleaving-effect-mixing-it-up-boosts-learning/>)

4. Starring Topics

- a. In the history tab, we plan to now be able to start studying/exploring previously starred/saved topics.
- b. This is a quality-of-life feature we thought of adding just after the ITR0 deadline ended.

5. Share Feature

- a. A link system to share topics, courses and quizzes to friends for them to see and try.
- b. This change was implemented to preserve the group feel of studying and to stay tighter to our goal of a better study tool for ALL students, while also providing a next-generation tool that enhances collaboration and engagement. By integrating such features, we ensure that students can benefit from modern technology without losing the core values of group study teamwork, shared learning, and academic growth.

Concerns With Project Going Forward

Some concerns with the project currently -

- API keys being public or mismatching
 - We need to find a way to have secure keys while being able to share the project, the current proposed solution is to use a .env file and then keep track of it on a different platform which is not public.
 - Refreshing API keys can be time-consuming, especially when free service API keys expire frequently and need to be updated in multiple files. This process diverts valuable time from creative tasks and problem-solving. Using a .env file to store API keys can simplify key management, reducing unnecessary effort and ensuring a more efficient workflow.
- Account system

- We need to associate account history and topics with a specific user rather than relying solely on memory.
 - A proposed solution is to store this information in the Firestore database alongside other details recorded during the registration process. The email and password will remain in a separate, secure database, ensuring that passwords remain inaccessible to anyone. Additionally, we currently use a separate database to store university names, and we can extend this database to store account history as well.
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- GPT-API
 - The current AI landscape is evolving at an incredibly rapid pace, making it difficult to decide which large language model (LLM) to use. Almost every week, a new and improved LLM is released, often promising better performance, lower costs, or easier integration. This constant innovation creates both opportunities and challenges, while newer models may offer advantages, they also require time to evaluate, test, and integrate effectively into our workflow. Despite the overwhelming number of choices, we have decided to stick with ChatGPT for our project. One of the main reasons for this decision is the extensive ecosystem of resources, documentation, and community support that OpenAI has built around it. ChatGPT's API is well-documented, widely adopted, and provides robust performance, making it a reliable choice for our needs. Additionally, the availability of numerous integrations, plugins, and developer tools further reinforces its suitability for our project.
 - As we move forward, we will ensure that we stay informed about advancements in the LLM space. However, for now, ChatGPT remains our preferred choice due to its stability, reliability, and the wealth of resources that support its implementation.